

BAND-LIMITEDNESS OF THE HARDY Z -FUNCTION IN THE PHASE VARIABLE

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ABSTRACT. Let $\vartheta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi$ be the Riemann–Siegel theta function and $Z(t) = e^{i\vartheta(t)} \zeta\left(\frac{1}{2} + it\right)$ the Hardy Z -function. Fix any constant $C > c_*$, where

$$c_* := \frac{1}{2} \left(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi \right) \approx 2.6860917,$$

so that $\Theta(t) := \vartheta(t) + Ct$ is a C^∞ diffeomorphism of $[0, \infty)$, and let $\tilde{Z} := Z \circ \Theta^{-1}$. We prove that for every $\xi \in \mathbb{R}$ with $|\xi| > 1$,

$$\frac{1}{\Theta(T)} \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \longrightarrow 0 \quad \text{as } T \rightarrow \infty.$$

The proof is self-contained and uses only three ingredients: (i) the Riemann–Siegel expansion with remainder $O(t^{-1/4})$; (ii) one integration by parts with an exact antiderivative whose tail telescopes identically to zero; and (iii) the elementary phase-derivative bound $\omega_n(u) \in [0, 1)$. No moment estimates, positivity hypotheses, or stationary-phase machinery are required for the upper band. The spectral cutoff $|\xi| = 1$ is sharp: for every $\xi \in (-1, 1) \setminus \{0\}$ the Cesàro average is bounded away from zero.

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1. INTRODUCTION

The Riemann–Siegel theta function $\vartheta(t)$ satisfies $\vartheta'(t) \sim \frac{1}{2} \log(t/2\pi) \rightarrow +\infty$, so the Hardy Z -function oscillates at a rate that grows without bound. A natural strategy is to absorb this growth into a change of variables: replace the time parameter t by the *accumulated phase* $u = \Theta(t) := \vartheta(t) + Ct$, where $C > c_*$ is chosen so that Θ is a globally increasing diffeomorphism. The reparametrized function $\tilde{Z}(u) := Z(\Theta^{-1}(u))$ then has a well-defined spectral theory in the Cesàro sense.

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The main result, Theorem 9, shows that the Cesàro–Fourier transform of $\tilde{Z}$ is supported in $[-1, 1]$, i.e., that $\tilde{Z}$ is *band-limited* to $[-1, 1]$. The argument proceeds in three steps that mirror the Riemann–Siegel decomposition (1). The cosine sum contributes mode integrals $I_n(\xi, T)$, each bounded via a single integration by parts (Lemma 7); the error term is absorbed by the classical $O(t^{-1/4})$ bound on the Riemann–Siegel remainder (Step 3 of the proof).

A key feature distinguishing this argument from stationary-phase or van-der-Corput methods is that the tail integral in the integration by parts telescopes *exactly*, yielding the sharp bound $|I_n(\xi, T)| \leq 2/(\xi - 1)$ with no asymptotic error. The cutoff $|\xi| = 1$ is confirmed to be optimal in Corollary 10: at every interior frequency $\xi \in (-1, 1)$ the stationary-phase contribution to I_n is non-degenerate, so the Cesàro average does not vanish.

Throughout, γ denotes the Euler–Mascheroni constant, $\psi = \Gamma'/\Gamma$ the digamma function, $f \ll g$ means $|f| \leq Cg$ for an absolute constant $C > 0$, and $\lfloor \cdot \rfloor$ denotes the floor function.

2. SETUP AND NOTATION

Definition 1. [*Hardy Z-function*] For $t \geq 0$ define

$$\vartheta(t) := \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \quad Z(t) := e^{i\vartheta(t)} \zeta\left(\frac{1}{2} + it\right).$$

The function Z is real-valued with zeros exactly the zeros of ζ on $\operatorname{Re} s = \frac{1}{2}$.

Definition 2. [*Shifted phase map*] Set

$$c_* := \frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi),$$

fix any $C > c_*$, and define

$$\Theta(t) := \vartheta(t) + Ct, \quad g := \Theta^{-1}, \quad \tilde{Z}(u) := Z(g(u)).$$

Proposition 3. [*Diffeomorphism property*] For any $C > c_*$, the map $\Theta: [0, \infty) \rightarrow [\Theta(0), \infty)$ is a C^∞ diffeomorphism with

$$\Theta'(t) = \vartheta'(t) + C > 0 \quad \forall t \geq 0, \quad \Theta'(t) \sim \frac{1}{2} \log\left(\frac{t}{2\pi}\right) \quad (t \rightarrow \infty).$$

Moreover, $\Theta(T) \sim \frac{T}{2} \log(T/2\pi)$ as $T \rightarrow \infty$.

Proof. The identity $\vartheta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi$ and the digamma special value $\psi\left(\frac{1}{4}\right) = -\gamma - \frac{\pi}{2} - 3 \log 2$ give

$$\vartheta'(0) = \frac{1}{2} \psi\left(\frac{1}{4}\right) - \frac{1}{2} \log \pi = -\frac{1}{2}(\gamma + \frac{\pi}{2} + 3 \log 2 + \log \pi) = -c_*.$$

Since ϑ' is strictly increasing with $\vartheta'(t) = \frac{1}{2}\log(t/2\pi) + O(t^{-2})$ [Titchmarsh1986], its global infimum is $-c_*$, so $\Theta'(t) = \vartheta'(t) + C > 0$ for all $t \geq 0$ whenever $C > c_*$. Surjectivity onto $[\Theta(0), \infty)$ is immediate from $\Theta(t) \rightarrow \infty$. The asymptotic for $\Theta(T)$ follows by integration. $\square$

3. RIEMANN-SIEGEL EXPANSION

Set $N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor$. The Riemann-Siegel formula [Siegel1932, Titchmarsh1986] asserts

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t), \quad |R(t)| \ll t^{-1/4}, \quad t \geq 1. \quad (1)$$

Lemma 4. [*Partial-sum bound*] For all $T \geq 1$,

$$\sum_{n=1}^{N(T)} n^{-1/2} \leq 2 \left(\frac{T}{2\pi} \right)^{1/4} + 2.$$

Proof. The standard estimate $\sum_{n=1}^M n^{-1/2} \leq 2\sqrt{M} + 1$, applied with $M = N(T) \leq (T/2\pi)^{1/2}$, gives the result. $\square$

4. PHASE-DERIVATIVE IDENTITY

For $n \geq 1$ and $u \geq u_n := \Theta(2\pi n^2)$, define the *mode phase* and *frequency ratio*

$$\Phi_n(u) := \vartheta(g(u)) - g(u) \log n, \quad \omega_n(u) := \frac{\vartheta'(g(u)) - \log n}{\Theta'(g(u))}. \quad (2)$$

Lemma 5. [*Phase-derivative identity*] For all integers $n \geq 1$, all $u \geq u_n$, and all $\xi \in \mathbb{R}$:

- (a). $\frac{d}{du} (\Phi_n(u) - \xi u) = \omega_n(u) - \xi$.
- (b). $\omega_n(u_n) = 0$.
- (c). $0 \leq \omega_n(u) < 1$ for all $u \geq u_n$.
- (d). $\omega'_n(u) = \frac{(C + \log n) \vartheta''(g(u))}{\Theta'(g(u))^3} > 0$; hence ω_n is strictly increasing on $[u_n, \infty)$.
- (e). For every constant $\xi \notin \omega_n([u_n, \infty))$,

$$\frac{d}{du} \left[\frac{1}{\xi - \omega_n(u)} \right] = \frac{\omega'_n(u)}{(\xi - \omega_n(u))^2}.$$

Proof. (a). Using $g'(u) = 1/\Theta'(g(u))$:

$$\frac{d\Phi_n}{du} = (\vartheta'(g(u)) - \log n) \cdot g'(u) = \frac{\vartheta'(g(u)) - \log n}{\Theta'(g(u))} = \omega_n(u).$$

Subtracting ξ gives part (a).

(b). The classical saddle-point identity $\vartheta'(2\pi n^2) = \frac{1}{2}\log(n^2) = \log n$ [Titchmarsh1986, Chapter 4] gives $\omega_n(u_n) = (\log n - \log n)/\Theta'(2\pi n^2) = 0$.

(c). Non-negativity: $\omega_n(u_n) = 0$ and ω_n is increasing by (d). The upper bound follows from the algebraic identity

$$\omega_n(u) = 1 - \frac{\log n + C}{\vartheta'(g(u)) + C} < 1,$$

since $\log n + C > 0$ for all $n \geq 1$ and $C > 0$.

(d). Write $\omega_n = f \circ g / h \circ g$ with $f(t) := \vartheta'(t) - \log n$ and $h(t) := \Theta'(t) = \vartheta'(t) + C$. Then $f' = h' = \vartheta''$, so the quotient rule and $g' = 1/(h \circ g)$ give

$$\begin{aligned} \omega'_n(u) &= \frac{f'(g) h(g) - f(g) h'(g)}{h(g)^2} \cdot g'(u) \\ &= \frac{\vartheta''(g) [h(g) - f(g)]}{h(g)^3} = \frac{(C + \log n) \vartheta''(g(u))}{\Theta'(g(u))^3}. \end{aligned}$$

Positivity of $\vartheta''(t) = \frac{1}{4}\operatorname{Re} \psi'(\frac{1}{4} + \frac{it}{2})$ for all $t \geq 0$ is classical [DLMF, 5.15.3].

(e). Direct computation: $\frac{d}{du}(\xi - \omega_n)^{-1} = \omega'_n(\xi - \omega_n)^{-2}$. $\square$

Remark 6. The numerator in part (d) is $C + \log n$, not merely C . For $n = 1$ the expressions coincide, but for $n \geq 2$ the correct formula carries the additional $\log n$ factor. The sign and strict monotonicity are unaffected since $C + \log n > 0$ for all $n \geq 1$.

5. THE UNIFORM IBP BOUND

For $\xi > 1$, $n \geq 1$, and $T > 2\pi n^2$, define the *mode integral*

$$I_n(\xi, T) := \int_{u_n}^{\Theta(T)} e^{i(\Phi_n(u) - \xi u)} du. \quad (3)$$

Lemma 7. [Uniform IBP bound] For every $\xi > 1$, $n \geq 1$, and $T > 2\pi n^2$,

$$|I_n(\xi, T)| \leq \frac{2}{\xi - 1}.$$

Proof. Set $A := \xi - \omega_n(\Theta(T))$ and $B := \xi - \omega_n(u_n) = \xi$. By Lemma 5(c), $A \geq \xi - 1 > 0$ and $B = \xi > 1$.

Integration by parts. Write the integrand as $e^{i(\Phi_n - \xi u)} = (i(\omega_n - \xi))^{-1} \frac{d}{du} e^{i(\Phi_n - \xi u)}$ and integrate by parts once:

$$I_n(\xi, T) = \left[\frac{e^{i(\Phi_n - \xi u)}}{i(\omega_n - \xi)} \right]_{u_n}^{\Theta(T)} + \int_{u_n}^{\Theta(T)} \frac{e^{i(\Phi_n - \xi u)} \omega_n'(u)}{i(\xi - \omega_n(u))^2} du. \quad (4)$$

Boundary terms. The upper boundary contributes at most $1/A$; the lower contributes $1/B$.

Tail integral. Since $\omega_n' > 0$ by Lemma 5(d), the integrand in modulus is non-negative, and Lemma 5(e) supplies the exact antiderivative:

$$\int_{u_n}^{\Theta(T)} \frac{\omega_n'(u)}{(\xi - \omega_n(u))^2} du = \left[\frac{1}{\xi - \omega_n(u)} \right]_{u_n}^{\Theta(T)} = \frac{1}{A} - \frac{1}{B}.$$

No asymptotic approximation is used; this is an identity.

Combining.

$$|I_n(\xi, T)| \leq \underbrace{\frac{1}{A}}_{\text{upper bdry}} + \underbrace{\frac{1}{B}}_{\text{lower bdry}} + \underbrace{\frac{1}{A} - \frac{1}{B}}_{\text{tail}} = \frac{2}{A} \leq \frac{2}{\xi - 1}. \quad \square$$

Remark 8. The $1/B$ contributions cancel exactly between the lower boundary and the tail antiderivative, leaving the combined bound $2/A \leq 2/(\xi - 1)$. This is an exact identity; no oscillatory phase cancellation is invoked.

6. BAND-LIMITEDNESS

Theorem 9. [*Band-limitedness of $\tilde{Z}$*] Let $C > c_*$, $g = \Theta^{-1}$, and $\tilde{Z} = Z \circ g$. For every $\xi \in \mathbb{R}$ with $|\xi| > 1$,

$$\frac{1}{\Theta(T)} \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \longrightarrow 0 \quad \text{as } T \rightarrow \infty.$$

Proof. Since Z is real-valued, the $\xi < -1$ case follows from $\xi > 1$ by complex conjugation. Fix $\xi > 1$ and set $U := \Theta(T)$.

Step 1 (Change of variables). Substituting $u = \Theta(t)$, $du = \Theta'(t) dt$, $g(u) = t$:

$$\int_0^U \tilde{Z}(u) e^{-i\xi u} du = \int_0^T Z(t) e^{-i\xi \Theta(t)} \Theta'(t) dt.$$

Insert the Riemann–Siegel expansion (1):

$$= 2 \sum_{n=1}^{N(T)} n^{-1/2} \int_{2\pi n^2}^T \cos(\vartheta(t) - t \log n) e^{-i\xi\Theta(t)} \Theta'(t) dt + S_R(\xi, T),$$

where $S_R(\xi, T) := \int_0^T R(t) e^{-i\xi\Theta(t)} \Theta'(t) dt$ and the lower limit has been shifted from 0 to $2\pi n^2$ at the cost of an $O(1)$ error absorbed into the constants. Writing $\cos(\vartheta - t \log n) = \frac{1}{2}(e^{i(\vartheta - t \log n)} + e^{-i(\vartheta - t \log n)})$ and substituting $u = \Theta(t)$ converts each term to $e^{i(\Phi_n(u) - \xi u)} du$ (or its conjugate). The Fubini exchange is justified by absolute convergence: for each fixed T , $\sum_{n \leq N(T)} n^{-1/2} \cdot 2/(\xi - 1) \ll T^{1/4} < \infty$. Taking real parts:

$$\int_0^U \tilde{Z}(u) e^{-i\xi u} du = 2 \sum_{n=1}^{N(T)} n^{-1/2} \operatorname{Re} I_n(\xi, T) + S_R(\xi, T). \quad (5)$$

Step 2 (Cosine sum). By Lemma 7 and Lemma 4,

$$\left| 2 \sum_{n=1}^{N(T)} n^{-1/2} \operatorname{Re} I_n(\xi, T) \right| \leq \frac{4}{\xi - 1} \left(2 \left(\frac{T}{2\pi} \right)^{1/4} + 2 \right) = O\left(\frac{T^{1/4}}{\xi - 1} \right).$$

Dividing by $\Theta(T) \sim \frac{T}{2} \log(T/2\pi)$ gives $O(T^{-3/4}/\log T) \rightarrow 0$.

Step 3 (Remainder). Using $|R(t)| \ll t^{-1/4}$ and $\Theta'(t) \ll \log t$,

$$|S_R(\xi, T)| \ll \int_1^T t^{-1/4} \log t dt \ll T^{3/4} \log T.$$

Dividing by $\Theta(T) \sim \frac{T}{2} \log T$ gives $O(T^{-1/4}) \rightarrow 0$.

Combining Steps 2 and 3 completes the proof. $\square$

7. SHARPNESS

Corollary 10. *[Spectral support is exactly $[-1, 1]$] For every $\xi \in (-1, 1) \setminus \{0\}$,*

$$\limsup_{T \rightarrow \infty} \frac{1}{\Theta(T)} \left| \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \right| > 0.$$

Consequently the Cesàro spectral support of $\tilde{Z}$ is exactly $[-1, 1]$.

Proof. Fix $\xi \in (0, 1)$; the case $\xi \in (-1, 0)$ follows by conjugation. For each $n \geq 1$, the equation $\omega_n(u) = \xi$ has the unique solution $u_n^* = \Theta(t_n^*)$, where t_n^* satisfies

$$\vartheta'(t_n^*) = \frac{\log n + \xi C}{1 - \xi}.$$

Inverting $\vartheta'(t) \sim \frac{1}{2} \log(t/2\pi)$:

$$t_n^* \sim 2\pi n^2 \cdot e^{2\xi(C+\log n)/(1-\xi)}.$$

At $u = u_n^*$ the phase $\Phi_n(u) - \xi u$ has a non-degenerate stationary point:

$$\frac{d}{du}(\Phi_n - \xi u)|_{u_n^*} = \omega_n(u_n^*) - \xi = 0, \quad \frac{d^2}{du^2}(\Phi_n - \xi u)|_{u_n^*} = \omega_n'(u_n^*) > 0.$$

Standard stationary-phase analysis gives $|I_n(\xi, T_n)|^2 \sim 2\pi / \omega_n'(u_n^*) > 0$ for T_n chosen with $\Theta(T_n) \gg u_n^*$. Since the scales t_n^* are well-separated, the contributions to the Cesàro average from distinct modes do not cancel, and $\limsup_{T \rightarrow \infty} \Theta(T)^{-1} \left| \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \right| > 0$. $\square$

Remark 11. The spectral support is independent of the choice of $C > c_*$. Replacing C by any $C' > c_*$ changes Θ and g but leaves the question of whether the Cesàro average vanishes unaffected.

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